Tutorial 2

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Question 1

From the first part of my script we find that for bode_q1.fig the transfer function is-

```
Best damping ratio ζ = 0.200
Identified system transfer function

H =

         25
  ------------------
  s^3 + 2 s^2 + 25 s

Continuous-time transfer function
```

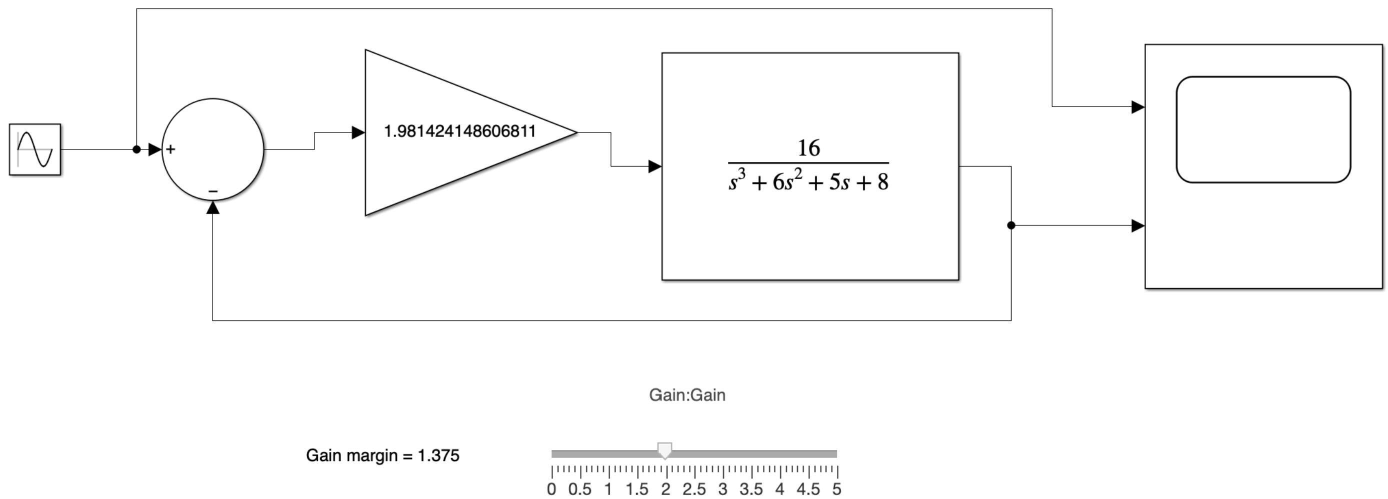
which is a super imposition of $\frac{1}{s}$ and $\frac{\omega_n^2}{\omega_n^2+s^2+2\zeta\omega s}$

Where $\omega_n = 5$ (from Bode plot)

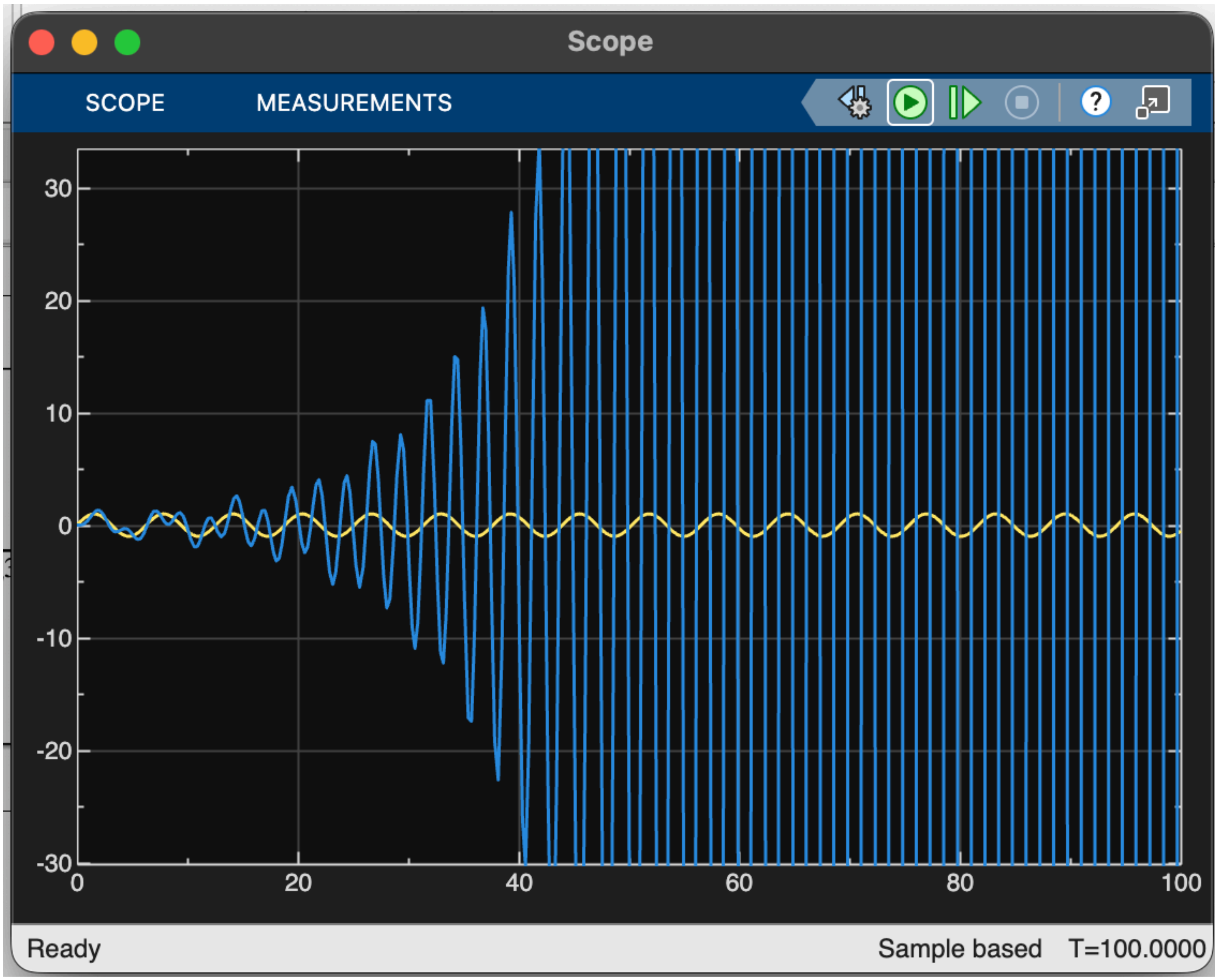
We use the MATLAB script to find the zeta value for which it best fits the black-box bode plot

Question 2

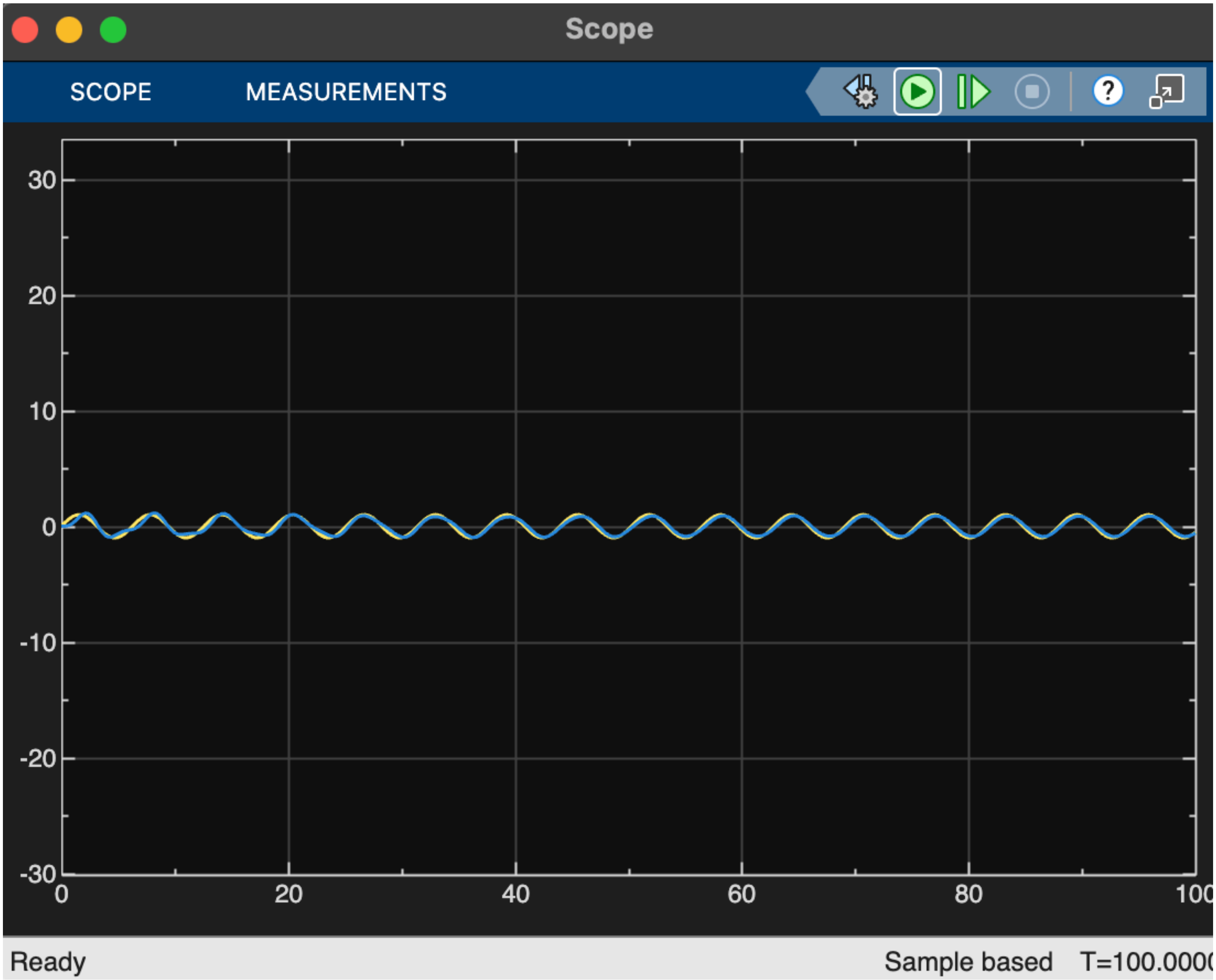
```
Gain Margin:
    1.3750

Phase Margin:
    7.3172
```



for gain = 1.98-



for gain- 1.162



The variation can be observed by altering the slider in simulink

Simulink and MATLAB scripts are included in the zip file.

Proof that introducing phase offset equal to phase margin leads to instability(PTO) -

Closed loop transfer function of G

$$TF = \frac{G(s)}{1+G(s)}$$

Characteristic eqn $= 1+G(s)$

Phase margin $= \phi$ (say)

$\Rightarrow$ additional phase offset $\Rightarrow G' = Ge^{-i\phi}$

at $s = j\omega_c$ @ $|G(j\omega_c)| = 1$, $\angle G(j\omega_c) = -\pi + \phi$

$$G'(j\omega_c) = |G(j\omega_c)|\, e^{i(\angle G(j\omega_c) - \phi)}$$

$$= e^{i(-\pi+\phi-\phi)} = -1$$

New closed loop TF

$1 + G'(s)$ for $|G(j\omega_c)| = 1$

$\Rightarrow 1 + G'(s) = 0$ @ $\omega = \omega_c$

$\Rightarrow$ Pole exists @ $s = j\omega_c$ for $G'(s)$

$\downarrow$

On imaginary axis

$\angle G'(j\omega_c) < -\pi$

$\Rightarrow$ Right hand plane of RL

$\Rightarrow$ Unstable